

**BOARDINGHUB: A SECURE WEB-BASED BOARDING HOUSE MANAGEMENT
SYSTEM WITH AUTOMATED BILLING AND EMAIL NOTIFICATION FOR
TENANTS**

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

This chapter presents the introduction of the study, which discusses the background of boarding house management and the problems encountered in its traditional setup. It also includes the objectives of the study, the significance of the study, the scope and limitation, the review of related literature and studies, and the definition of terms. These sections collectively establish the foundation of the research and provide a clear understanding of the need to develop a secure web-based boarding house management system.

Introduction

Boarding houses provide affordable housing for students and workers who live far from their homes, especially near universities and workplaces. However, many boarding house owners manually manage tenant information and payment records using notebooks, paper records, or simple spreadsheets. This traditional approach often leads to misplaced records, incorrect data entries, and difficulty tracking tenant payments. As the number of tenants increases, manual record management becomes more complex and time-consuming. Another challenge is the lack of an organized communication system between landlords and tenants. Many landlords rely on verbal reminders, text messages, or social media to inform tenants about payment deadlines and announcements. Without a notification system, tenants may forget about their payment schedules, which can lead to late payments.

Recent studies have attempted to improve property and boarding house management through digital platforms. Saputra et al. (2025) and Kartinah (2025) developed systems using the Laravel framework to automate room management and registration. Mangca (2023) and Bautista (2025) integrated notification and occupancy tracking systems, while Cholaraja et al. (2024) developed the “Justrent” system to improve rent collection and financial reporting. However, these systems still have limitations. Many focus primarily on room booking and lack

automated billing features. Others rely on SMS notifications, which can be unreliable and do not provide formal documentation for financial transactions. Additionally, most systems do not integrate with commonly used payment platforms in the Philippines, particularly GCash, which requires landlords to manually verify payments and issue receipts. Given these limitations, there is still a need for a unified and secure system that integrates digital payments, automated billing, and formal email-based documentation.

To address the issue, this study proposes BoardingHub: A Secure Web-Based Boarding House Management System with Automated Billing and Email Notification for Tenants. The system provides a centralized digital platform that simplifies boarding house management by allowing landlords to manage tenant records, track room occupancy, and track payments. One of the key features of BoardingHub is its automated billing system, which generates monthly billing records and sends payment notifications to tenants via email. The system also integrates a GCash payment allowing tenants to conveniently pay their rent online. Once the payment is confirmed, the system automatically sends an email notification and receipt to the tenant as confirmation of the transaction. In addition, BoardingHub emphasizes data security by implementing user authentication and role-based access control (RBAC) to protect tenant information and financial records. Built with PHP/Laravel and MySQL, the system aims to transform the traditional manual management of boarding houses into a more efficient and secure digital system.

Objective of the Study

This study aims to develop a Secure Web-Based Boarding House Management System for tenants to the City of Ilagan.

Specifically, this study seeks to:

1. Determine the problems involved in the existing system of manual boarding house management in terms of tenant records, billing, and payment monitoring.
2. Develop a Boarding House Management System that contains the following modules:

2.1 Dashboard

For Landlord:

- 2.1.a View overall tenant list
- 2.1.b Monitor payment status of all tenants (paid, unpaid, partial, overdue)
- 2.1.c View total collected payments
- 2.1.d Receive payment notifications
- 2.1.e View list of tenants with overdue balances
- 2.1.f View recent transactions (latest payments made)

For Tenant:

- 2.1.g View payment history
- 2.1.h View billing details
- 2.1.i Receive payment reminders/notification
- 2.1.j Check payment status (paid, unpaid, partial)

2.2 Login and Security Module

- 2.2.a User login (landlord and tenant)
- 2.2.b Password encryption
- 2.2.c Role-based access control
- 2.2.d Session management (login/logout)

2.3 Tenant Management Module

- 2.3.a Add tenant information
- 2.3.b Update tenant records

2.3.c Delete tenant records

2.3.d Assign rooms to tenants

2.4 Billing Module

2.4.a Generate billing statements

2.4.b Set due dates

2.4.c Calculate total rent and balances

2.4.d View billing history

2.5 Payment Monitoring Module

2.5.a Record payments

2.5.b Track payment status (paid/unpaid/partial/overdue)

2.5.c Monitor outstanding balances

2.5.d View payment transactions

2.6 Email Notification Module

2.6.a Send payment reminders

2.6.b Send payment confirmation notifications

2.6.c Notify tenants of due dates

2.6.d Notify landlord of received payments

2.7. e Landlord post and send announcements

2.7 Receipt Generation Module

2.7.a Generate digital receipts

2.7.b Include payment details (amount, date, tenant name)

2.7.c Send receipt via email

2.7.d Store receipt records

3. Test the developed system through comprehensive testing procedures, including:
 - 4.1 Functionality Testing
 - 4.2 Usability Testing
 - 4.3 Security Testing
 - 4.4 User Acceptance Testing (UAT)
4. Evaluate the developed system using ISO 25010 in terms of:
 - 4.1 Functional Suitability
 - 4.2 Usability
 - 4.3 Security
 - 4.4 Reliability
5. To integrate Gcash payment, allowing tenants to pay conveniently and securely.
6. To apply basic security measures such as authentication, access control, and data protection to ensure the safety of user information.

Scope and Limitation

This study focuses on the design, development, and evaluation of **BoardingHub: A Secure Web-Based Boarding House Management System** to enhance the efficiency of managing tenant information, billing, and payment monitoring for landlords and tenants. It examines the process of constructing a system that organizes tenant records, automates billing, sends email notifications, and integrates GCash payment. The study also explores how manual and traditional methods of managing tenant information, billing, and payments affect the accuracy, organization, and convenience of boarding house operations.

Login and Security Module ensures that the system is accessed only by authorized users. It allows landlords and tenants to log in securely using their credentials. The module includes password encryption to protect sensitive data, role-based access control (RBAC) to restrict access based on user roles, and session management to maintain secure login sessions and prevent unauthorized access.

Tenant Management Module enables landlords to manage tenant records efficiently. This includes adding new tenant information, updating existing records, deleting tenant data when necessary, and assigning rooms to tenants. This module ensures that all tenant and occupancy information is properly organized and easily accessible.

Billing Module automates the creation and management of billing statements. It allows landlords to generate monthly bills, set due dates, calculate rent and other charges, and monitor balances. The module also maintains a history of all billing records, enabling both landlords and tenants to review previous transactions.

Payment Monitoring Module is responsible for recording and tracking all payment transactions. It allows landlords to record payments made by tenants and automatically updates their payment status. The system identifies whether payments are complete, partial, unpaid, or overdue, and provides a clear overview of outstanding balances and transaction history.

Email Notification Module enhances communication by sending automated messages to users. Tenants receive notifications such as payment reminders, due date alerts, payment confirmations, and announcements from landlords. Landlords also receive notifications when payments are made. This reduces the need for manual communication and ensures timely information delivery.

Receipt Generation Module automatically creates digital receipts after a successful payment. Each receipt contains important details such as the tenant's name, payment amount,

and transaction date. The system sends the receipt via email and stores it in the database for future reference and documentation.

In addition, the system integrates GCash payment to allow tenants to pay rent online in a convenient and secure manner. Once the payment is verified, the system automatically updates the payment status and sends a confirmation email with the corresponding receipt.

Limitation

Despite its capabilities, the system has certain limitations. BoardingHub is designed only as a web-based system and does not include a mobile application. It also requires a stable internet connection to function and cannot operate offline.

The system supports only GCash as its primary online payment method and does not fully integrate other payment platforms such as PayMaya or bank transfers beyond basic functionality.

Advanced security features such as biometric authentication, multi-factor authentication (MFA), and blockchain-based security are not included in the system. The system relies only on standard security measures such as password encryption and role-based access control.

The evaluation of the system is limited to selected landlords and tenants in the City of Ilagan. Therefore, the results may not fully represent all boarding house management scenarios in other locations or larger-scale operations.

Significance of the Study

This study will be beneficial to the following:

Landlords. The proposed system will help landlords efficiently manage tenant information, billing, and payment monitoring. It reduces manual work, minimizes errors, and

improves organization of records. The system also enables landlords to receive timely payment updates and notifications, making management more convenient and effective.

Tenants. The system will provide tenants with an easy and convenient way to view their billing information, payment history, and account status. It also allows them to receive email notifications and digital receipts, improving communication and ensuring transparency in their transactions.

Students. This study will serve as a learning resource for students, particularly those in Information Technology and related fields. It provides knowledge on web-based system development, database management, online payment integration, and implementation of basic security features.

Instructors. The study can be used as a reference material for teaching system development, programming, and software design. It can help instructors demonstrate real-world applications of web-based systems and enhance students' understanding of system integration and development processes.

Future Researchers. The findings and design of this study may serve as a reference for future research related to boarding house management systems, web-based applications, and automated billing and notification systems. It can also be a basis for further improvements and innovations.

Review of Related Literature and Studies

Web-Based Boarding House Management Systems

The transition from manual to digital management in boarding house operations has been extensively documented in recent literature, with researchers consistently identifying the limitations of paper-based record keeping and the potential of web-based platforms to streamline administrative workflows. Saputra et al. (2025) designed and developed a web-based room management information system for Pak Yadi Boarding House, aimed at automating and streamlining key administrative tasks. Their system demonstrated that digital transformation could effectively organize tenant records into accessible databases, though the implementation focused primarily on room allocation without integrated billing automation or payment gateway connectivity. This gap between administrative and financial functions highlights the need for more comprehensive systems that unify operational management with payment processing.

Similarly, (Kartinah, 2025) designed and developed a website-based application to facilitate online registration for Supiati Boarding House, emphasizing user-friendly interfaces for landlords to monitor occupancy status and maintain tenant records digitally. While representing progress from paper-based management, the study acknowledged limitations in payment processing capabilities, as the system did not incorporate automated billing cycles or digital payment collection mechanisms. Mangca (2023) addressed communication challenges by developing a booking system using the Laravel framework with integrated SMS notification capabilities, recognizing that timely communication regarding payment deadlines and room availability was critical for operational efficiency.

Bautista (2025) extended functionality by designing a web-based dormitory management system for Cagayan State University-Aparri aimed at modernizing the management of boarding house facilities. An iterative Agile framework was adopted to allow

for continuous refinement based on user feedback during implementation. The system integrated an advanced room allocation feature that simplified the process of managing various housing units and student profiles. The application automated the entire invoicing workflow, significantly reducing the time spent on manual ledger entries and financial reconciliation. The research identified that while notification systems improved communication efficiency, reliance on manual payment verification and lack of automated receipt generation continued to create administrative bottlenecks.

Nugraha et al. (2025) employed Agile methodology in creating the SIMKO boarding house management system using Laravel, emphasizing iterative development cycles that incorporated continuous feedback from boarding house operators. Despite methodological improvements, the study noted that payment integration remained limited to manual recording of cash transactions, indicating that even recently developed systems have not fully addressed digital payment needs. Puspasari et al. (2025) focused on information accessibility for both landlords and prospective tenants, emphasizing transparent information sharing in rental transactions. The absence of automated billing generation and secure payment processing within the same platform created discontinuities in the tenant experience that required switching to separate channels for financial transactions.

Billing Automation and Financial Management Systems

The automation of billing processes represents a critical advancement in property management technology, addressing persistent challenges of manual calculation errors, delayed invoice generation, and inconsistent payment tracking. Mobile payments and web portals provide extended customer interaction by presenting customers with real-time billing details and transaction records. Auto Bill also supports contactless billing, which is in line with health and safety needs of today's world. Kshirsagar et al. (2025) developed an AI-based automatic billing system demonstrating how intelligent algorithms could transform traditional

billing workflows by automatically calculating charges, generating invoices without human intervention, and identifying patterns that might indicate billing anomalies. Their research established that artificial intelligence integration could reduce processing time by up to 70% while improving accuracy rates, suggesting that similar automation approaches could benefit boarding house environments where recurring monthly rent calculations follow predictable patterns.

Encarnacion et al. (2025) specifically addressed the Philippine context by designing an online payment monitoring system for student boarding houses near university campuses. Their study recognized that student tenants often faced challenges tracking payment deadlines across academic schedules, leading to late payments and administrative complications. The system implemented real-time payment status tracking that allowed both tenants and landlords to view current balances, payment histories, and upcoming due dates through a unified interface.

Surbakti et al. (2025) approached financial management through mobile technology, developing an Android-based financial management information system using the Rapid Application Development method. Their research acknowledged that many boarding house operators managed finances through informal record-keeping that created difficulties in tracking revenue and identifying overdue accounts. The mobile application provided landlords with portable access to financial data, enabling immediate recording of payments and real-time consultation of tenant account statuses.

Khaliq et al. (2025) designed a boarding house management information system with explicit focus on administration and financial reporting features, recognizing that effective property management requires both operational efficiency and financial transparency. Their system architecture separated administrative functions from financial modules while maintaining data consistency across these domains through integrated database design. Despite

these advances, the study acknowledged that the system relied on manual payment recording rather than automated payment gateway integration, creating potential delays between tenant payment actions and system record updates. Group, Goel, and colleagues (2022) examined flexible rent payment options from the tenant perspective, analyzing how payment flexibility influenced tenant satisfaction and retention. Their research established that tenants increasingly expected digital payment methods that accommodated varying income schedules, particularly among student and young professional demographics that constitute significant portions of boarding house populations.

Notification Systems and Communication Technology in Property Management

Effective communication between landlords and tenants constitutes a fundamental operational requirement in boarding house management, yet traditional methods often prove unreliable or inefficient. Fernandez (2022) developed a dormitory management information system that integrated SMS notification capabilities with biometric security features, addressing both communication and access control challenges in student housing environments. However, the study noted that SMS notifications lacked the formal documentation trail that email systems provide, and the cost of SMS delivery at scale could become prohibitive for larger boarding house operations, suggesting that email-based notification systems might offer more sustainable alternatives for comprehensive documentation.

Nesta et al. (2025) employed the Rapid Application Development method to design a web-based boarding house management information system that prioritized rapid deployment and iterative refinement based on user feedback. The RAD approach enabled quick implementation of core features including tenant registration, room assignment, and notification capabilities, with subsequent iterations adding functionality based on actual usage patterns. The study emphasized that notification systems must balance comprehensiveness with

user preferences, as excessive notifications could lead to alert fatigue while insufficient communication left tenants uninformed about critical matters such as payment deadlines. Magno et al. (2024) developed an apartment rental management system focused on real-time transaction processing and task organization, implementing instant notification triggers that alerted landlords when payments were recorded and notified tenants when maintenance requests were updated. The real-time architecture ensured that information asymmetries between landlords and tenants were minimized, reducing disputes arising from outdated or conflicting information.

Digital Payment Integration in Housing Management Systems

The integration of digital payment processing into property management systems addresses both convenience and security concerns in financial transactions. Wardana et al. (2026) developed a web-based boarding house management information system with explicit payment gateway integration, recognizing that modern tenants expected digital payment convenience in rent transactions. Their system architecture incorporated secure payment processing that allowed tenants to complete rent payments through integrated interfaces rather than separate banking applications or cash transactions. The research demonstrated that payment gateway integration reduced the administrative burden of manual payment recording and reconciliation, while providing tenants with immediate transaction confirmation that improved satisfaction and trust.

Tangsripiroj et al. (2025) advanced integrated housing management by developing Dormlander, a smart dormitory management system combining utility metering with digital payment solutions. Their research recognized that boarding house billing often involved variable utility charges in addition to fixed rent amounts, creating complexity that manual systems handled poorly. By integrating automated utility metering with digital payment

processing, the system ensured that tenants received accurate bills reflecting actual consumption and could immediately settle accounts through the same platform.

Cholaraja et al. (2024) developed Justrent as a comprehensive house rental management system that improved rent collection and financial reporting through digital platform implementation. Their research addressed the broader rental property market, establishing system architectures that could potentially adapt to boarding house contexts. The study demonstrated that integrated financial management combining rent collection, expense tracking, and reporting generation provided landlords with unprecedented visibility into business operations, enabling strategic decisions based on actual performance data rather than informal estimates.

Security Architecture and Access Control in Management Systems

The protection of sensitive tenant and financial information in boarding house management systems requires robust security architectures extending beyond basic authentication mechanisms. Harun et al. (2022) developed a tenant rental system using the CodeIgniter framework and Bulma CSS framework that emphasized security considerations including password encryption, session management, and input validation to protect against common web vulnerabilities. Their implementation demonstrated that even lightweight frameworks could achieve adequate security when properly configured, though the study acknowledged that more sophisticated access control mechanisms such as role-based authorization would enhance security in multi-user environments.

Pratama et al. (2025) extended boarding house system functionality by developing a web-based information system with GPS integration specifically designed for student housing applications. Their research recognized that location-based services could enhance

management by enabling proximity-based features, though GPS integration required careful attention to privacy protection and data security as location information constitutes sensitive personal data. Habib et al. (2023) examined boarding house management enhancement through Customer Relationship Management integration, analyzing how relationship management principles could improve operational outcomes. Their research established that systematic management of tenant relationships through digital platforms improved retention rates and reduced vacancy periods by enabling personalized communication and service delivery.

Theoretical Foundations: Digital Payment Systems and Financial Technology

The theoretical understanding of digital payment adoption and implementation provides essential context for boarding house management system development. Agarwal et al. (2024) conducted a comprehensive review of digital payment systems in India, identifying current trends and future directions that inform system design decisions in similar emerging market contexts. Mohan et al. (2024) developed a framework for assessing online payment systems in small businesses, recognizing that smaller enterprises such as boarding house operations face unique challenges in payment system adoption including limited technical expertise and resource constraints.

Tierney et al. (2023) examined secure online payment processing, identifying issues and challenges in modern retailing that apply equally to rental payment contexts. Their analysis emphasized that security concerns represent primary barriers to digital payment adoption, requiring systems to implement robust encryption, authentication, and fraud prevention measures.

Subramanian et al. (2024) analyzed FinTech innovations and their impact on traditional banking, identifying challenges and prospects for integrated financial services. Their research highlighted how non-bank platforms such as property management systems could leverage

financial technology to offer enhanced services beyond traditional banking relationships. Tan et al. (2023) studied mobile payment technologies, identifying current trends and future directions that inform the selection of payment gateway technologies for boarding house systems. Their analysis established that mobile-first payment solutions dominate emerging markets, supporting the prioritization of GCash integration for Philippine boarding house contexts.

Yadav et al. (2024) examined the role of payment gateways in e-commerce success, establishing that payment system reliability and user experience significantly influence transaction completion rates and customer satisfaction. Their research suggested that boarding house management systems should prioritize payment gateway selection based on reliability, security, and user familiarity rather than solely on transaction cost considerations. Wang et al. (2023) analyzed factors influencing mobile payment adoption in emerging markets, identifying trust, perceived usefulness, and ease of use as critical determinants of user acceptance. These findings inform the design of payment interfaces within boarding house systems, emphasizing the importance of transparent transaction processes and clear confirmation mechanisms.

Cloud Computing and System Infrastructure

The infrastructure supporting web-based management systems significantly influences their reliability, scalability, and security characteristics. Chen et al. (2024) examined the impact of digital transformation on small enterprises through case study approaches, identifying how cloud-based systems enable smaller organizations to access enterprise-grade technology capabilities without proportional infrastructure investments. Their research established that cloud computing levels the competitive playing field for small boarding house operators, enabling them to offer digital services previously available only to larger property management companies.

El-Masri et al. (2023) reviewed challenges of cloud computing in higher education, identifying issues of data sovereignty, vendor lock-in, and connectivity dependency that inform infrastructure decisions for boarding house systems. Their analysis suggested that hybrid cloud approaches might offer optimal balance between accessibility and control for sensitive tenant and financial data. O'Reilly et al. (2023) examined cloud computing in educational management, identifying benefits and challenges that parallel boarding house operational contexts including multi-user access, data security requirements, and cost management considerations.

Xu et al. (2024) analyzed trends in cloud computing for small business management, reviewing how cloud-based systems support remote access, automatic updates, and disaster recovery capabilities that are particularly valuable for boarding house operators who may lack dedicated IT support staff. Patel et al. (2024) developed scalable solutions for payment processing in high-traffic scenarios, establishing architectural principles that ensure payment systems remain responsive during peak transaction periods such as monthly rent collection cycles.

User Experience and System Adoption

The successful implementation of boarding house management systems depends fundamentally on user acceptance and effective interaction design. Hall et al. (2023) conducted a comprehensive review of user experience in mobile payment systems, identifying design principles that promote user confidence and transaction completion. Their research established that clear feedback, transparent processes, and error prevention mechanisms significantly influence user satisfaction with digital payment interfaces. Woods et al. (2023) examined the effect of user experience on payment systems in mobile applications, finding that interface design quality directly impacts payment system adoption and continued usage.

Bock et al. (2023) analyzed social media use in community management, identifying communication patterns and preferences that inform notification system design for boarding house platforms. Their research suggested that tenants expect timely, relevant communications delivered through their preferred channels, supporting the implementation of email notification systems that provide formal documentation while accommodating individual preferences. Khmer et al. (2024) examined mobile applications in property management, identifying trends toward feature integration and challenges of maintaining simplicity while expanding functionality.

Security and Privacy in Digital Payment Systems

The protection of sensitive financial and personal information represents a critical requirement for boarding house management systems. Rao et al. (2024) navigated challenges of online payment security, identifying threat vectors and protective measures essential for secure payment processing. Their research emphasized the importance of encryption, secure authentication, and regular security audits in maintaining payment system integrity. Rehman et al. (2023) developed a framework for e-governance in property management, identifying integration challenges and solutions that apply to digital transformation in rental housing contexts.

Singh et al. (2023) analyzed trends and challenges of e-commerce in supply chain management, identifying principles of transaction transparency and audit trail maintenance that inform payment recording requirements in boarding house systems. Thakur et al. (2023) reviewed potential of blockchain technology in e-commerce, identifying opportunities for enhanced transaction security and transparency that may inform future boarding house system evolution. Zhang et al. (2023) examined the impact of e-commerce on traditional retail through

case study approaches, establishing analytical frameworks for assessing digital transformation outcomes that apply to boarding house management modernization.

Synthesis and Research Gap

The reviewed literature and studies reveal significant progress in digital boarding house management while identifying persistent gaps that the proposed BoardingHub system addresses. Existing systems have successfully implemented tenant information management, room occupancy tracking, and basic notification capabilities, yet consistently lack integration between automated billing generation, email-based formal documentation, and Philippine-specific digital payment platforms. Security implementations in current systems typically rely on basic authentication without comprehensive role-based access control architectures that differentiate landlord and tenant capabilities while protecting sensitive data.

The theoretical foundations established by digital payment research provide robust frameworks for implementing secure, user-friendly payment integration, while cloud computing literature offers guidance on scalable, reliable infrastructure deployment. User experience research emphasizes the importance of transparent processes and clear feedback mechanisms that build tenant confidence in digital payment systems. The convergence of these technological capabilities with the specific operational requirements of Philippine boarding house management creates the opportunity for BoardingHub to advance beyond existing solutions by providing unified platforms that automate billing, integrate GCash payment processing, deliver formal email documentation, and implement secure role-based access control within a single coherent system architecture specifically designed for the City of Ilagan context.

Definition of Terms

The following terms used in the study are defined operationally to ensure clarity and better understanding.

Automated Billing System – A feature that generates billing statements without manual input. Its effectiveness is measured by comparing generated bills with expected computations and due dates.

BoardingHub – The web-based system developed in this study. Its effectiveness will be measured through functionality, usability, security, and user acceptance, using surveys, observation, and system testing.

Database Management – Refers to how the system stores, retrieves, and updates data. It is evaluated by checking data accuracy, consistency, and retrieval speed during system use.

Functionality Testing – The process of checking whether system features work as intended. It is measured by executing test cases and recording if outputs match expected results.

ISO 25010 Model – A software quality evaluation standard used in this study. It is applied by measuring system performance based on functional suitability, usability, reliability, and security through survey questionnaires.

Password Encryption – The process of converting user passwords into secure coded formats. Its effectiveness is measured by verifying that stored passwords in the database are unreadable and protected against unauthorized access.

Role-Based Access Control (RBAC) – Refers to a security mechanism where system access is restricted based on user roles (landlord or tenant). In this study, it is measured by testing whether users can only access features assigned to their role during system testing.

Security Testing – The process of identifying system vulnerabilities. It is measured by testing login security, data protection, and resistance to unauthorized access.

Session Management – Refers to the system's ability to maintain and control user login sessions. It is evaluated by testing automatic logout, session expiration, and prevention of unauthorized session reuse.

Usability Testing – Refers to evaluating how easy and user-friendly the system is. It is measured using user surveys and task completion rates.

User Acceptance Testing (UAT) – A testing process where actual users evaluate the system. It is measured through survey results using ISO 25010 criteria such as usability, functionality, reliability, and security.

CHAPTER 2

This chapter presents the conceptual framework, system architecture, context diagram, data flow diagram, entity relationship diagram, database schema, and Gantt chart of the system

TECHNICAL BACKGROUND

Conceptual Framework

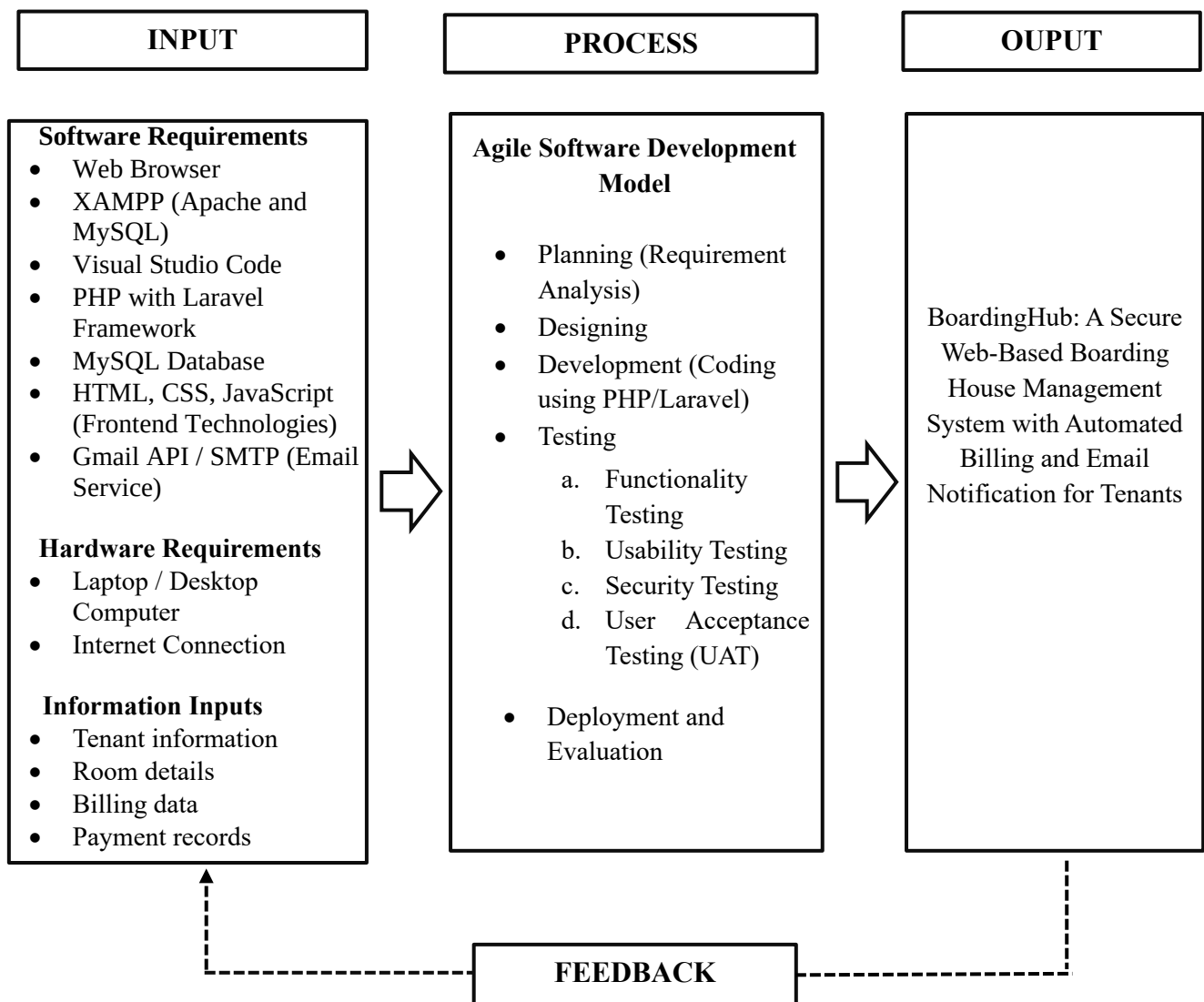


Figure 2-1. *Conceptual Framework of Boarding House Management System using the Input–Process–Output (IPO) model*

Figure 2.1 presents the conceptual framework of the proposed system, BoardingHub: A Secure Web-Based Boarding House Management System with Automated Billing and Email Notification for Tenants, using the Input–Process–Output (IPO) model.

The **Input** phase includes both software and hardware requirements necessary for the development of the system. Software requirements consist of a web browser, PHP with Laravel framework, XAMPP, Visual Studio Code, MySQL database, and front-end technologies such as HTML, CSS, and JavaScript, along with Gmail API/SMTP for email services. Hardware requirements include a laptop or desktop computer and a stable internet connection to support system development and usage. It also includes the information inputs that consists of tenant information, such as personal details and room assignments; room data, including availability and rental rates; billing information, such as monthly charges and due dates; and payment records, including transaction details and payment status.

The **Process** phase follows the Agile Software Development Model, which involves iterative steps such as planning (requirements analysis), designing, development (coding using PHP/Laravel), and testing. The testing phase includes functionality testing, usability testing, security testing, and user acceptance testing (UAT) to ensure the system meets user needs and quality standards. Deployment and evaluation are also conducted to finalize the system and assess its performance.

The **Output** of the process is a secure web-based boarding house management system that automates tenant management, billing generation, payment monitoring, and email notification. This system aims to improve efficiency, accuracy, and communication between landlords and tenants.

The **Feedback** component is important because it allows continuous improvement of the system. Feedback is collected from users, such as landlords and tenants, during testing and evaluation. Their suggestions and experiences are used to refine the system, fix errors, and improve its performance.

System Architecture

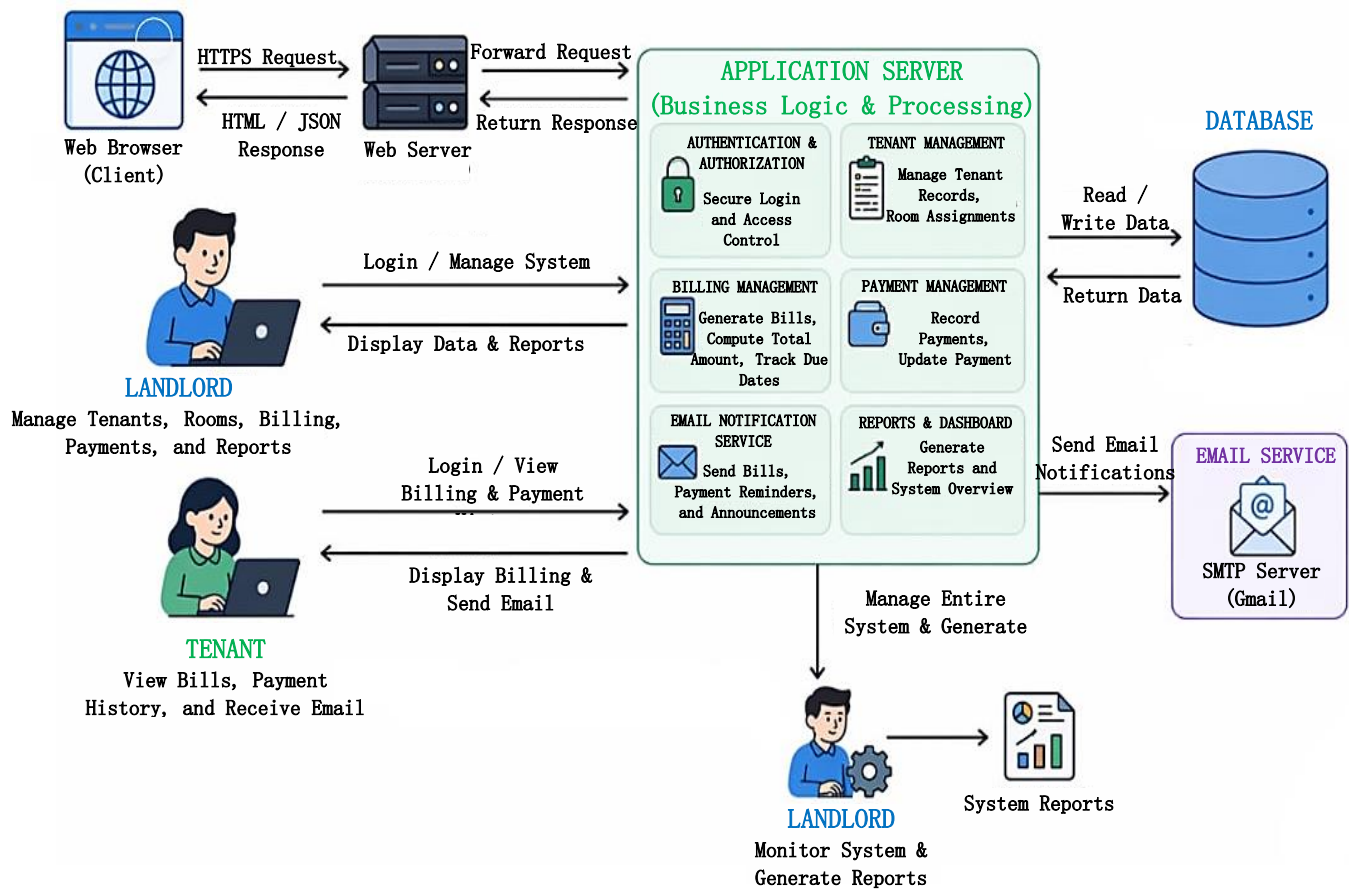


Figure 2-2. System Architecture of Boarding House Management System

Figure 2.2 illustrates the three-layer architecture of the Boarding House Management System, consisting of the presentation layer, application layer, and data layer. The presentation layer provides the web-based interface used by landlords and tenants to access the system. The application layer serves as the core processing unit, handling essential business logic such as, authentication and authorization, tenant and room management, billing generation, payment processing, email notifications, and report generation. The data layer is responsible for storing and managing all operational data, including tenant records, billing information, and payment history within a centralized database. Additionally, external services such as email delivery systems support communication by sending billing reminders and system notifications, ensuring efficient and secure interaction between users and the system.

Context Diagram

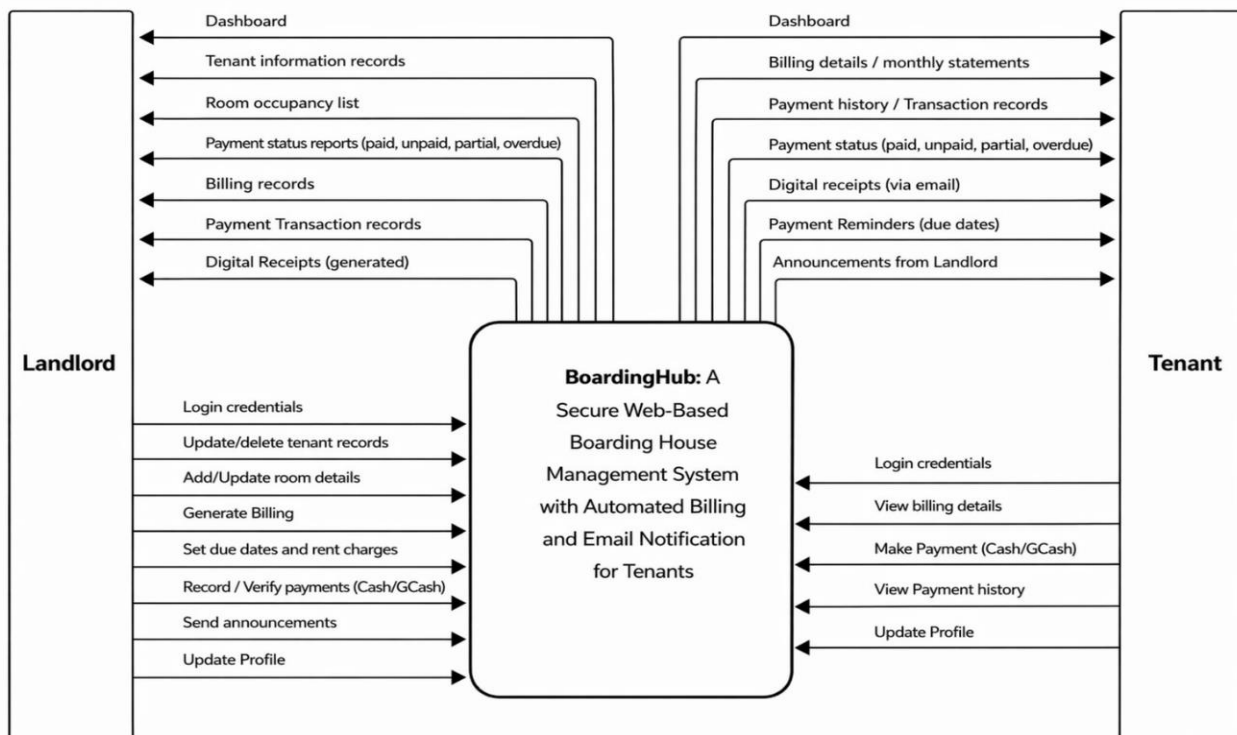


Figure 2-3. Context Diagram of Boarding House Management System

Figure 2.3 presents the high-level interaction between the BoardingHub system and external entities, primarily the landlord and tenant users. The landlord uses the system to handle important tasks such as managing tenant records, updating room details, generating billing, setting due dates, verifying payments, and sending announcements. On the other hand, tenants use the system to view their billing details, make payments, check their payment history, receive digital receipts and reminders, and update their profiles. All of this information is processed by the system, which then provides useful outputs like dashboards, payment status updates, transaction records, and notifications. Overall, the diagram shows how data flows between the landlord, tenant, and the system, as well as the main functions of the system.

Data Flow Diagram

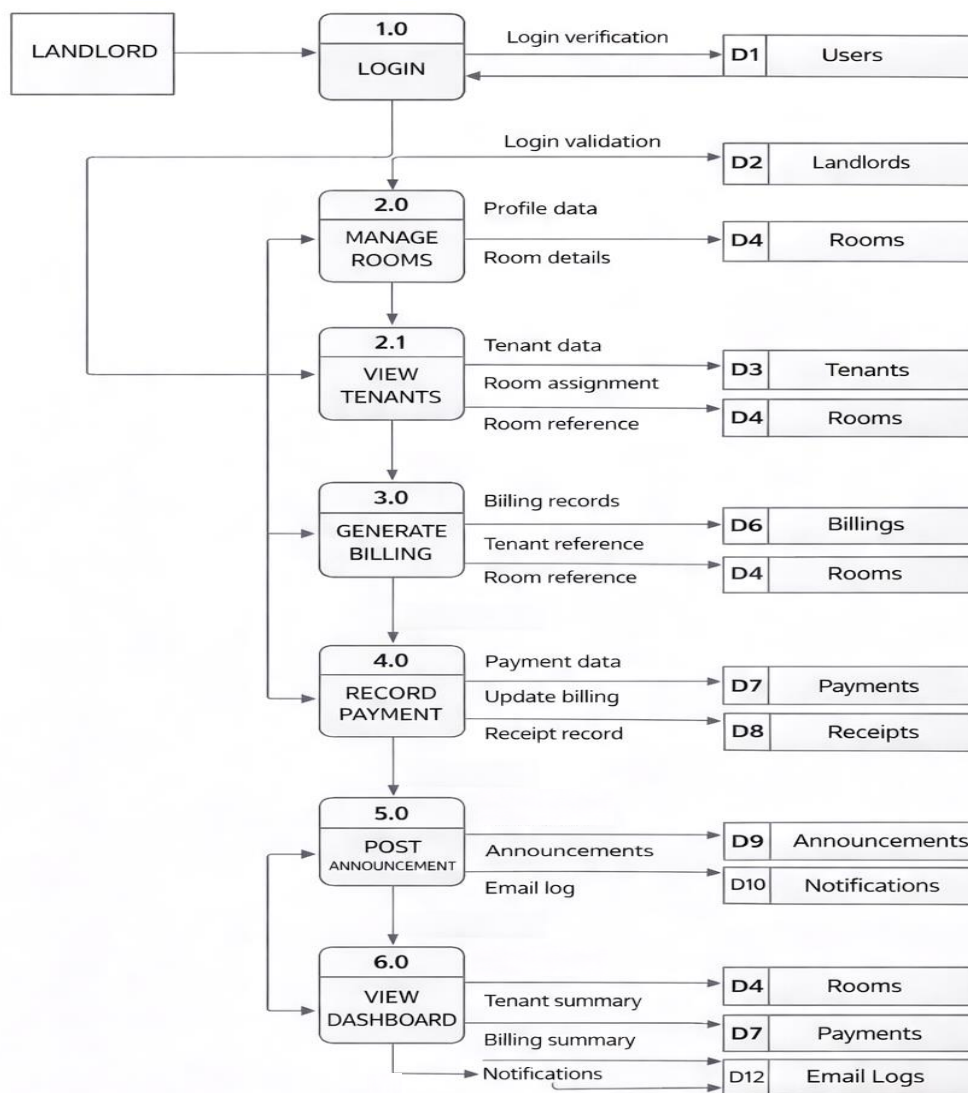


Figure 2-4. Data Flow diagram of Landlord

The Data Flow Diagram (DFD) of the Landlord shows how the landlord interacts with the Boarding House Management System and how information moves within the system. The process begins with Login, where the landlord enters credentials and the system verifies them using the Users data store. After successful login, the landlord can manage rooms, where room details and availability are stored and updated in the Rooms data store. The landlord can also view tenant information, which is retrieved from the Tenants data store. The system then allows the landlord to generate billing records for tenants, which are saved in the Billings data store. When tenants make payments, the landlord records them through the Record Payment process,

where payment details and receipts are stored in the Payments and Receipts data stores. The landlord can also post announcements to inform tenants about important updates, which are saved in the Announcements data store and logged for notifications. Lastly, the landlord can access the dashboard, which displays a summary of important information such as rooms, tenants, billing, and payment records.

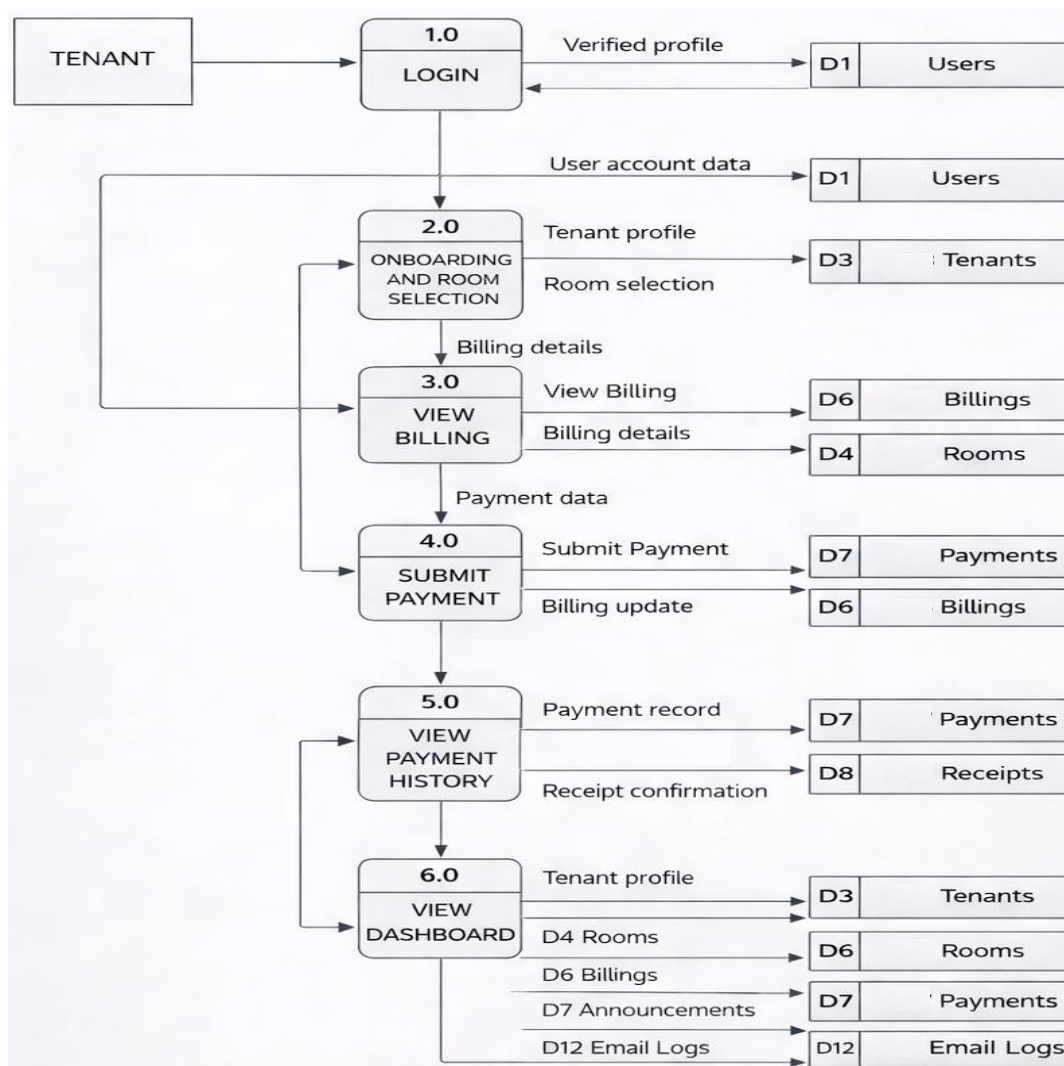


Figure 2-4. Data Flow diagram of Tenant

The Data Flow Diagram (DFD) of the Tenant illustrates how tenants interact with the Boarding House Management System. The process starts with Login, where the tenant enters login credentials or registers in the system, and the information is verified through the Users data

store. After logging in, the tenant proceeds to Onboarding and Room Selection, where tenant profile information is stored in the Tenants data store and available rooms are retrieved from the Rooms data store. The tenant can then view billing information, which is obtained from the Billings data store. If the tenant decides to pay the bill, they proceed to Submit Payment, where payment details are recorded in the Payments data store and the billing record is updated. The tenant can also view payment history, which retrieves payment records and receipts from the Payments and Receipts data stores. Finally, the tenant can access the dashboard, which provides an overview of the tenant's profile, current room assignment, billing summary, payments, announcements, and notifications. This allows tenants to easily monitor their account and transactions within the system.

Use Case Diagram

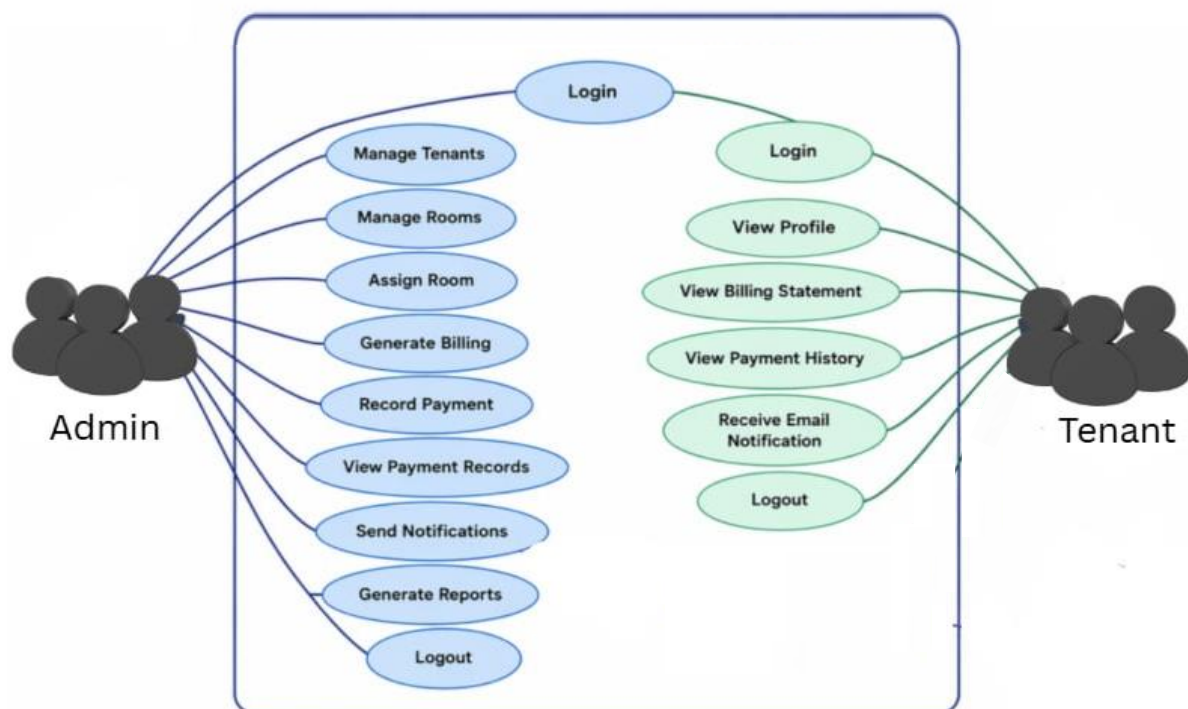


Figure 2.5. Use Case Diagram of Boarding House Management System

Figure 2.5 presents the functional interactions of two primary actors: landlord and tenant. Landlord use cases include managing rooms, viewing tenants, generating billings, monitoring

payments, and viewing reports. Tenant use cases include account access, onboarding, room selection, viewing bills, submitting payments, and viewing receipt/payment history. The diagram clarifies role-specific responsibilities and system access boundaries.

Entity Relationship Diagram

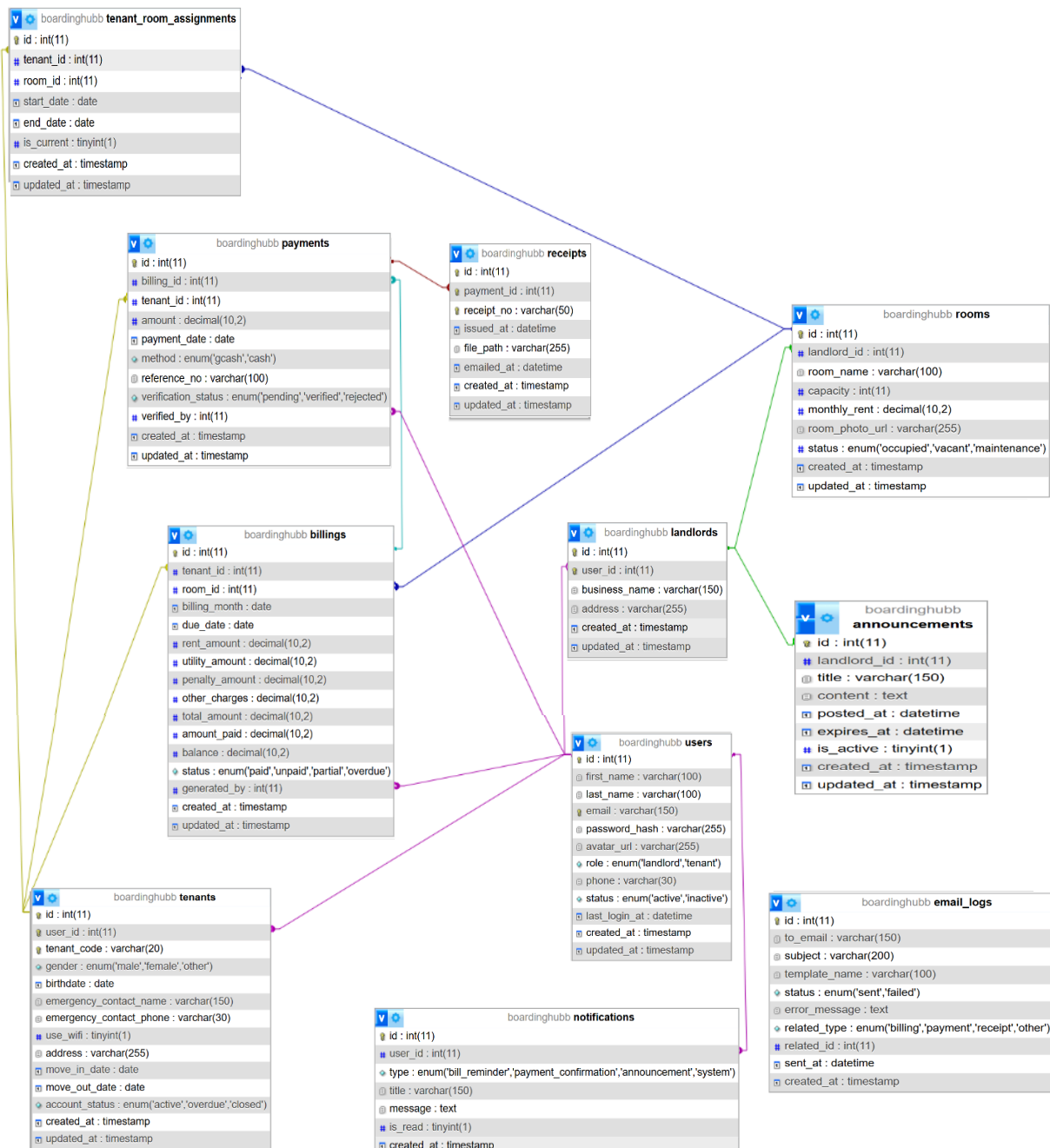


Figure 2.6. Entity Relationship Diagram (ERD) of Boarding House Management System

Figure 2.6 describes the logical database structure of BoardingHub. Core entities include users, landlords, tenants, rooms, tenant-room assignments, billings, payments, receipts, notifications, announcements, and email logs. The ERD defines primary and foreign key relationships to maintain data integrity and support accurate occupancy tracking, billing computation, and payment monitoring.

Database Schema

Schema for Users

Table 2.1 users

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
	first_name	VARCHAR(100)	Sherly
	last_name	VARCHAR(100)	Talaga
	email	VARCHAR(150)	talagasherly@gmail.com
	password_hash	VARCHAR(255)	\$2y\$10\$.....
	role	ENUM('landlord','tenant')	tenant
	auth_provider	ENUM('google')	google
	google_sub	VARCHAR(100)	1082...
	phone	VARCHAR(20)	09171234567
	status	ENUM('active','inactive')	active
	last_login_at	DATETIME	2026-04-18 10:30:00
	created_at	TIMESTAMP	2026-01-15 08:00:00
	updated_at	TIMESTAMP	2026-04-18 10:30:00

Table 2.1 stores every login account: name, email, role, Google identity, phone, and account status. It is the parent table for landlords and tenants.

Schema for Tenants

Table 2.2 tenants

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
FK	user_id	INT(11)	5
	tenant_code	VARCHAR(20)	TNT-0001
	gender	ENUM('male','female')	female
	birthdate	DATE	2000-05-10
	emergency_contact_name	VARCHAR(150)	Maria Santos
	emergency_contact_phone	VARCHAR(20)	09981234567
	use_wifi	TINYINT(1)	1
	move_in_date	DATE	2026-02-01
	move_out_date	DATE	NULL
	address	VARCHAR(255)	Ilagan City, Isabela
	created_at	TIMESTAMP	2026-02-01 09:00:00
	updated_at	TIMESTAMP	2026-04-01 14:00:00

Table 2.2 holds tenant-specific profile and occupancy-related fields linked one-to-one to users.

Schema for Landlords

Table 2.3 landlords

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
FK	user_id	INT(11)	3
	business_name	VARCHAR(150)	BoardingHub Main
	address	VARCHAR(255)	City of Ilagan, Isabela
	created_at	TIMESTAMP	2026-01-10 08:00:00
	updated_at	TIMESTAMP	2026-04-10 16:00:00

Table 2.3 stores landlord business details and extends the users row for the landlord role.

Schema for Rooms

Table 2.4 rooms

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
FK	landlord_id	INT(11)	1
	room_name	VARCHAR(100)	Room A-101
	capacity	INT(11)	4
	monthly_rent	DECIMAL(10,2)	3500.00
	room_photo_url	VARCHAR(255)	https://.../room.jpg
	status	ENUM('occupied','vacant','maintenance')	occupied
	created_at	TIMESTAMP	2026-01-12 10:00:00
	updated_at	TIMESTAMP	2026-03-20 11:00:00

Table 2.4 lists each rentable room, its capacity, rent, photo, and availability status under a landlord.

Schema for Tenant Room Assignments

Table 2.5 tenant_room_assignments

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
FK	tenant_id	INT(11)	1
FK	room_id	INT(11)	1
	start_date	DATE	2026-02-01
	end_date	DATE	NULL
	is_current	TINYINT(1)	1
	created_at	TIMESTAMP	2026-02-01 09:30:00
	updated_at	TIMESTAMP	2026-04-01 08:00:00

Table 2.5 links tenants to rooms over time and marks the current assignment for onboarding and capacity checks.

Schema for Billings

Table 2.6 billings

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
FK	tenant_id	INT(11)	1
FK	room_id	INT(11)	1
	billing_month	DATE	2026-04-01
	due_date	DATE	2026-04-15
	rent_amount	DECIMAL(10,2)	3500.00
	utility_amount	DECIMAL(10,2)	500.00
	other_charges	DECIMAL(10,2)	0.00
	total_amount	DECIMAL(10,2)	4000.00
	amount_paid	DECIMAL(10,2)	2000.00
	balance	DECIMAL(10,2)	2000.00
	status	ENUM('paid','unpaid','partial','overdue')	partial
FK	generated_by	INT(11)	3
	created_at	TIMESTAMP	2026-04-01 08:00:00
	updated_at	TIMESTAMP	2026-04-10 12:00:00

Table 2.6 stores monthly billing lines, charge breakdown, running balance, and billing status per tenant and room.

Schema for Payments

Table 2.7 payments

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
FK	billing_id	INT(11)	1
FK	tenant_id	INT(11)	1
	amount	DECIMAL(10,2)	2000.00
	payment_date	DATE	2026-04-10
	method	ENUM('gcash','cash')	gcash
	reference_no	VARCHAR(100)	GCA123456789
	verification_status	ENUM('pending','verified','rejected')	verified
FK	verified_by	INT(11)	3

Key	Field Name	Data Type	Example
	created_at	TIMESTAMP	2026-04-10 14:00:00
	updated_at	TIMESTAMP	2026-04-10 14:00:00

Table 2.7 records each payment against a billing, including method, reference, and verification state.

Schema for Receipts

Table 2.8 receipts

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
FK	payment_id	INT(11)	1
	receipt_no	VARCHAR(50)	RCP-20260410-000001
	issued_at	DATETIME	2026-04-10 14:05:00
	file_path	VARCHAR(255)	NULL
	emailed_at	DATETIME	NULL
	created_at	TIMESTAMP	2026-04-10 14:05:00
	updated_at	TIMESTAMP	2026-04-10 14:05:00

Table 2.8 stores one receipt per verified payment for proof and history display.

Schema for Announcements

Table 2.9 announcements

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
FK	landlord_id	INT(11)	1
	title	VARCHAR(150)	Water interruption
	content	TEXT	Supply will resume...
	posted_at	DATETIME	2026-04-12 09:00:00
	expires_at	DATETIME	2026-04-20 23:59:59
	created_at	TIMESTAMP	2026-04-12 09:00:00
	updated_at	TIMESTAMP	2026-04-12 09:00:00

Table 2.9 holds landlord announcements shown to tenants, with optional expiry.

Schema for Notifications

Table 2.10 notifications

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
FK	user_id	INT(11)	5
	type	ENUM('bill_reminder','payment_confirmation','announcement','system')	payment_confirmation
	title	VARCHAR(150)	Payment received
	message	TEXT	Your payment was recorded...
	is_read	TINYINT(1)	0
	created_at	TIMESTAMP	2026-04-10 14:06:00

Table 2.10 stores in-app notices for users (bills, payments, announcements, system).

Schema for Email Logs

Table 2.11 email_logs

Key	Field Name	Data Type	Example
PK	id	INT(11)	1
	to_email	VARCHAR(150)	tenant@gmail.com
	subject	VARCHAR(200)	Payment Receipt
	template_name	VARCHAR(100)	NULL
	status	ENUM('sent','failed')	sent
	error_message	TEXT	NULL
	related_type	ENUM('billing','payment','receipt','other')	receipt
	related_id	INT(11)	1
	sent_at	DATETIME	2026-04-10 14:07:00
	created_at	TIMESTAMP	2026-04-10 14:07:00

Table 2.11 records outbound email attempts and outcomes for auditing and troubleshooting.

GANTT CHART

This Gantt Chart presents the weekly activities of the researchers in developing the BoardingHub system. It covers the planning, design, development, testing, and documentation phases of the study.

Table 3-1. *Activities for January*

Activities	Week 1	Week 2	Week 3	Week 4
Selection of group mates				
Title formulation				
Conceptualization of study				

Table 3-2. *Activities for February*

Activities	Week 1	Week 2	Week 3	Week 4
Finalization of title				
Review of Related Literature				
Identify the problem				
Chapter I – Introduction				
Objectives of the Study				
Scope and Limitation				

Table 3-3. *Activities for March*

Activities	Week 1	Week 2	Week 3	Week 4
Review of Related Literature (finalization)				
Definition of Terms				
Chapter II – Technical Background				
Conceptual Framework				
System Architecture				
Context Diagram				
Data Flow Diagram				
Entity Relationship Diagram				

Table 3-4. *Activities for April*

Activities	Week 1	Week 2	Week 3	Week 4
Meeting with adviser				
Chapter I & II finalization				
Chapter III – Methodology				

CHAPTER 3

METHODOLOGY

This chapter presents the methodology used in developing and evaluating Boarding House Management System. It detailed the research design, Agile development process, respondents, data gathering methods, research instruments, and statistical tools. These procedures ensured that the resulting system was systematically developed and properly evaluated for quality and user acceptability.

Research Design

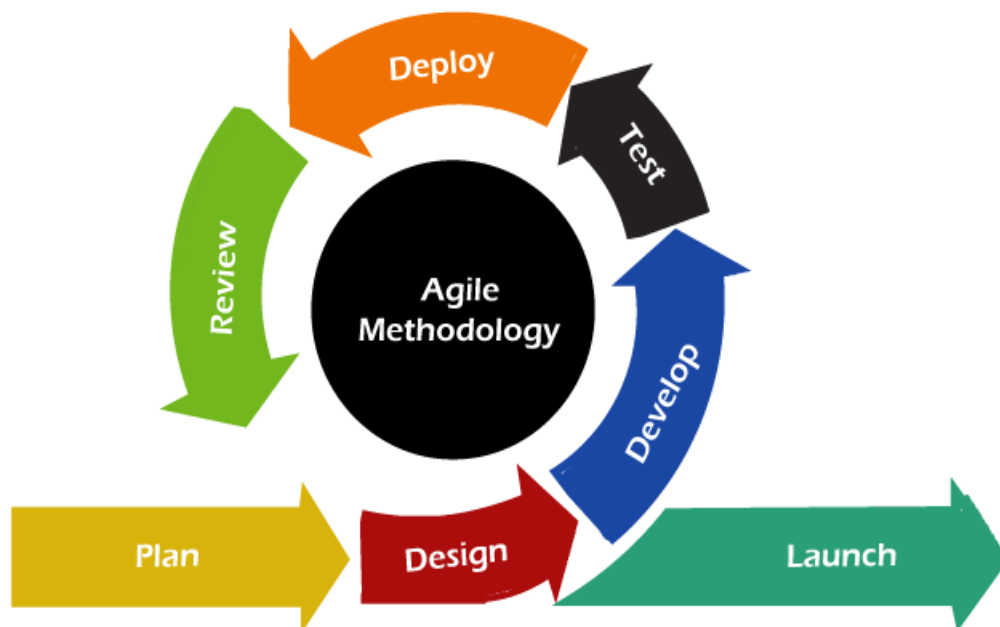


Figure 2-8. *Agile Software Development Methodology*

This study will utilize the **Agile Software Development Methodology**, which is an iterative and incremental approach to system development. Agile allows continuous improvement of the system through repeated cycles of planning, designing, developing, testing, and evaluating. The Agile model is appropriate for this study because it enables flexibility, faster feedback, and continuous enhancement of the Boarding House Management System. The researchers will follow the Agile development process which includes the following phases:

Planning Phase

In this phase, the researchers will identify the problems in manual boarding house management such as disorganized records, delayed payment monitoring, and lack of notification systems. The researchers will also determine the system requirements, features, and security needs of the proposed system.

Design Phase

The researchers will design the system structure including the database design, system architecture, and user interface. System diagrams such as: Context Diagram, Data Flow Diagram (DFD), Use Case Diagram and Entity Relationship Diagram (ERD) will be used as guides in developing the system. Security measures such as authentication and access control will also be included in the design.

Development Phase

In this phase, the actual coding of the system will be done. The researchers will develop the modules including:

- Login and security module
- Tenant management module
- Billing module
- Payment monitoring module
- Email notification module
- Receipt generation module

The system will be developed using the following software and hardware requirements:

Software Requirements

- Web Browser
- XAMPP (Apache Server and MySQL environment)

- Visual Studio Code (Source Code Editor)
- PHP with Laravel Framework
- MySQL Database
- HTML, CSS, JavaScript (Frontend Technologies)
- Gmail API / SMTP (Email Service)

Hardware Requirements

- Laptop / Desktop Computer
- Stable Internet Connection

Testing Phase

The developed system will undergo testing to ensure that all functions work correctly. Errors found during testing will be fixed and improvements will be applied. The following tests will be conducted:

- Functionality Testing
- Usability Testing
- Security Testing
- User Acceptance Testing (UAT)

Evaluation Phase

Selected users such as landlords and tenants will test the system and provide feedback to determine if the objectives of the study were achieved. The system will be evaluated using ISO 25010 in terms of:

- Functional Suitability
- Usability
- Security
- Reliability

Respondents of the Study

The respondents of this study consist of thirty-five (35) individuals, composed of five (5) boarding house landlords and thirty (30) tenants from the City of Ilagan who are directly involved in boarding house operations and management.

Landlords serve as the main administrators of the system. They are responsible for managing tenant records, generating billing statements, monitoring payments, and sending notifications. Their participation is essential in evaluating the system's functionality, usability, and effectiveness in improving boarding house management.

Tenants, on the other hand, are the end-users of the system. They will use the platform to view billing details, track payment history, receive notifications, and access digital receipts. Their feedback is important in assessing the system's usability, convenience, and overall user experience.

The selected respondents will evaluate the system based on the **ISO 25010 software quality standards**, particularly in terms of functional suitability, usability, reliability, and security. Their responses will provide valuable data in determining the effectiveness and acceptability of the proposed Boarding House Management System.

Data Gathering Procedure

The researchers will gather information through interviews, observation, and review of related literature to ensure a comprehensive understanding of boarding house management practices and system requirements.

1. Interview

Interviews will be conducted with selected boarding house owners to understand their current system, challenges, and needs. This method allows the researchers to obtain

detailed and first-hand information regarding how landlords manage tenant records, billing, payment monitoring, and communication with tenants.

2. Observation

The researchers will also conduct observation to examine how landlords manage tenant records, billing, and payments manually in actual practice. This method helps the researchers understand the real workflow of boarding house operations, including common difficulties, delays, and errors encountered in manual processes.

3. Review of Related Literature

A review of related literature and studies will be conducted to support the design and development of the system. Previous research, journals, and existing systems related to boarding house management will be analyzed to identify existing solutions, their limitations, and best practices. This process will help the researchers establish a strong theoretical foundation and guide the development of the proposed system features such as automated billing, payment monitoring, and notification systems.

Sampling Technique

The study will use **purposive sampling**, where participants are selected based on their relevance to the study. The respondents consisted of two (2) groups: boarding house landlords and tenants in the City of Ilagan. These groups were selected because they represent the primary users and beneficiaries of the BoardingHub system. By selecting participants who are actively involved in boarding house management and rental activities, the researchers can obtain accurate, relevant, and experience-based data. This approach strengthens the reliability of the findings and ensures that the results of the study are useful for the development, improvement, and evaluation of the proposed BoardingHub system, particularly in terms of functionality, usability, reliability, and security.

Research Instrument

The main instrument used in this study is a survey questionnaire based on the ISO 25010 Software Quality Model. The questionnaire is designed to evaluate the performance and quality of the proposed BoardingHub system in terms of Functional Suitability, Usability, Reliability, and Security. These criteria are used to determine how effectively the system meets user requirements and how well it performs its intended functions. The questionnaire will be distributed to the selected respondents, which include boarding house landlords and tenants.

A **Likert Scale** will be used to measure the responses of the participants. This scale represents the level of agreement or evaluation of the respondents regarding the system's performance.

SCALE	RANGE	INTERPRETATION
5	4.50 – 5.00	Highly Acceptable
4	3.50 – 4.49	Acceptable
3	2.50 – 3.49	Moderately Acceptable
2	1.50 – 2.49	Less Acceptable
1	1.00 – 1.49	Not Acceptable

Table 4-1. *Likert Scale*

Statistical Treatment

The data collected from the respondents will be analyzed using the following statistical tools to determine the effectiveness and acceptability of the proposed BoardingHub system.

1. Frequency and Percentage

Frequency and percentage will be used to summarize the number of responses for each category. This will determine the distribution of responses from the participants based on each evaluation criterion and provide a clear representation of user feedback.

Formula:

$$\text{Percentage} = \frac{f}{N} \times 100 \quad (1)$$

Where:

- f = frequency of responses
- N = total number of respondents

2. Weighted Mean

Weighted Mean will be used to determine the overall evaluation of the system based on the responses of the participants. It will measure the level of system performance in terms of Functional Suitability, Usability, Reliability, and Security.

Formula:

$$\bar{X} = \frac{\sum(f \cdot x)}{N} \quad (2)$$

Where:

- f = frequency of responses
- x = rating value
- N = total number of respondents

Interpretation Scale:

SCALE	RANGE	INTERPRETATION
5	4.50 – 5.00	The system fully meets user requirements and performs very well with no major issues.
4	3.50 – 4.49	The system meets user requirements and performs well with only minor issues.
3	2.50 – 3.49	The system partially meets user requirements but needs improvement in several areas.
2	1.50 – 2.49	The system has noticeable problems and does not fully satisfy user requirements.
1	1.00 – 1.49	The system fails to meet user requirements and performs poorly.

Table 4-2. *Interpretation Scale*

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